

ORF307

The Simplex method

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Today's topics

- Iterate between neighboring basic solutions
- Optimality conditions
- Simplex iterations

Recap

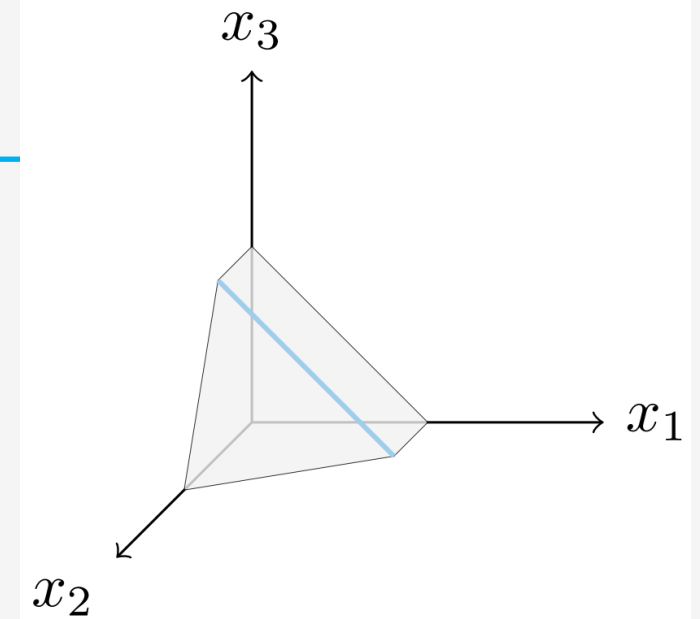
Constructing a basic feasible solution

$$\min \quad c^T y$$

$$\text{s.t.} \quad x_1 + x_3 = 1$$

$$(1/2)x_1 + x_2 + (1/2)x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$



Only $n - m = 1$ **inequality has to be active** (i.e., tight $x_i = 0, i = 1, 2, 3$)

You could set $x_1 = 0$ and solve

$$\begin{bmatrix} 1 & 0 & 1 \\ 1/2 & 1 & 1/2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 1/2 \end{bmatrix} \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$$

Setting $x_2 = 0$ or $x_3 = 0$ you get different basic solutions

Constructing basic solutions

A **basic solution** is computed by following the following steps:

Step #1: Choose any m independent columns of $A \rightarrow A_{B(1)}, \dots, A_{B(m)}$ (basic columns)

Step #2: Set $x_i = 0$ for all $i \neq B(1), \dots, B(m)$

Step #3: solve $Ax = b$ for the remaining variables $\rightarrow x_{B(1)}, \dots, x_{B(m)}$ (basic variables)

$$Ax = b \Leftrightarrow A_{\mathbf{B}}x_{\mathbf{B}} = b \Leftrightarrow \begin{bmatrix} | & & | \\ A_{B(1)} & \dots & A_{B(m)} \\ | & & | \end{bmatrix} \begin{pmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

If $x_{\mathbf{B}} \geq 0$ then x is a **basic feasible solution**

Standard form polyhedra

Consider the LP in standard form:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

Assumptions:

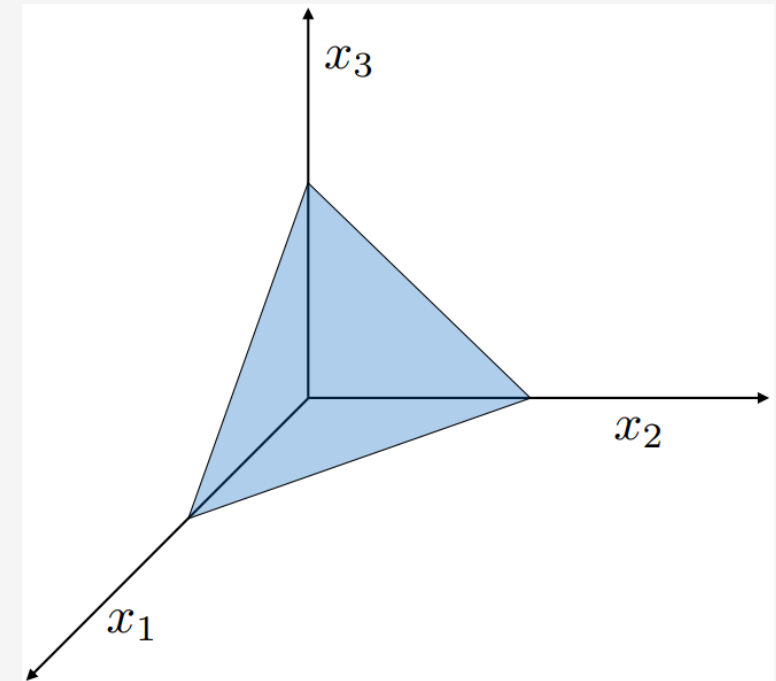
$A \in R^{m \times n}$ has full **row** rank and $m \leq n$

Interpretation:

P is an $(n - m)$ -dimensional surface

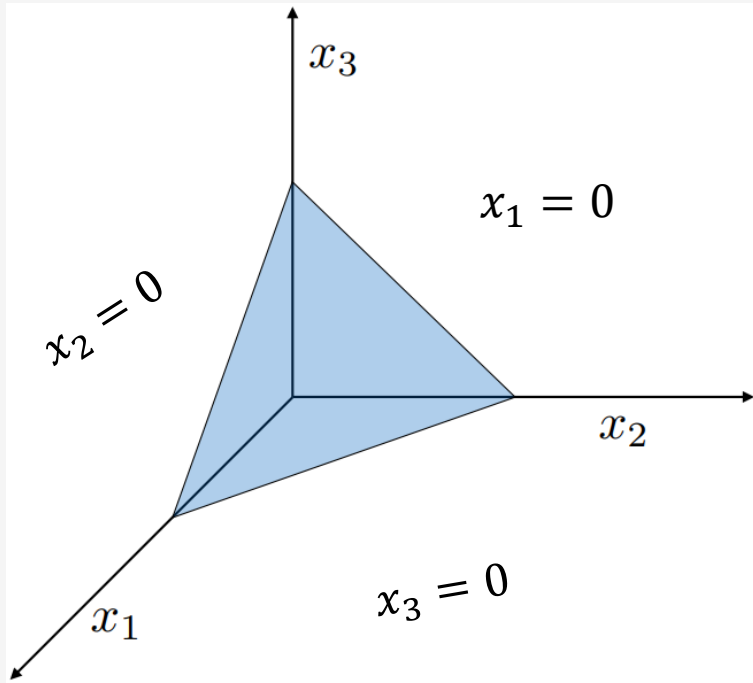
The feasible region is a standard polyhedron

$$P = \{x \in R^n \mid Ax = b, x \geq 0\}$$

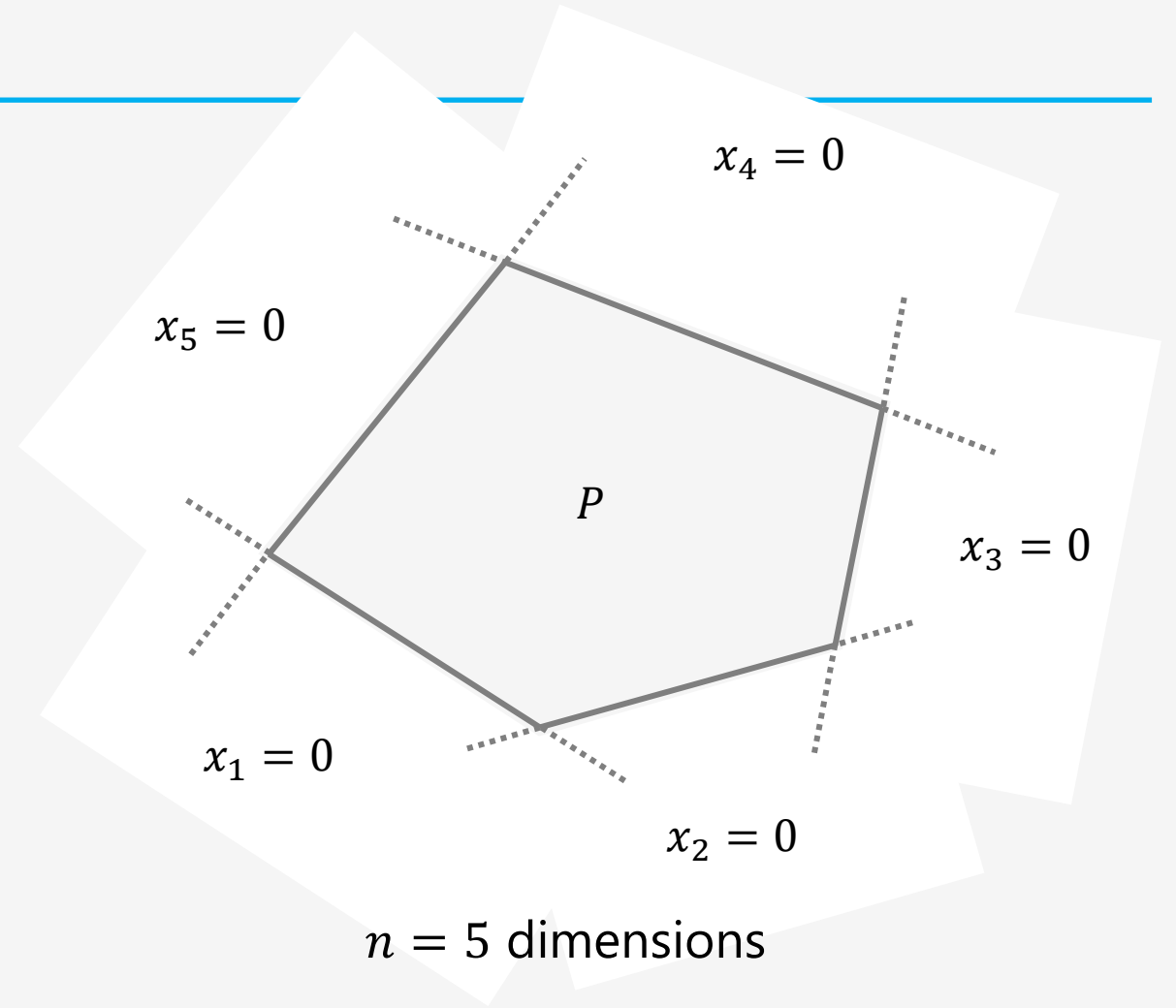


Example with $m = 1$ and $n = 3$

Standard polyhedra - visualization



$n = 3$ dimensions



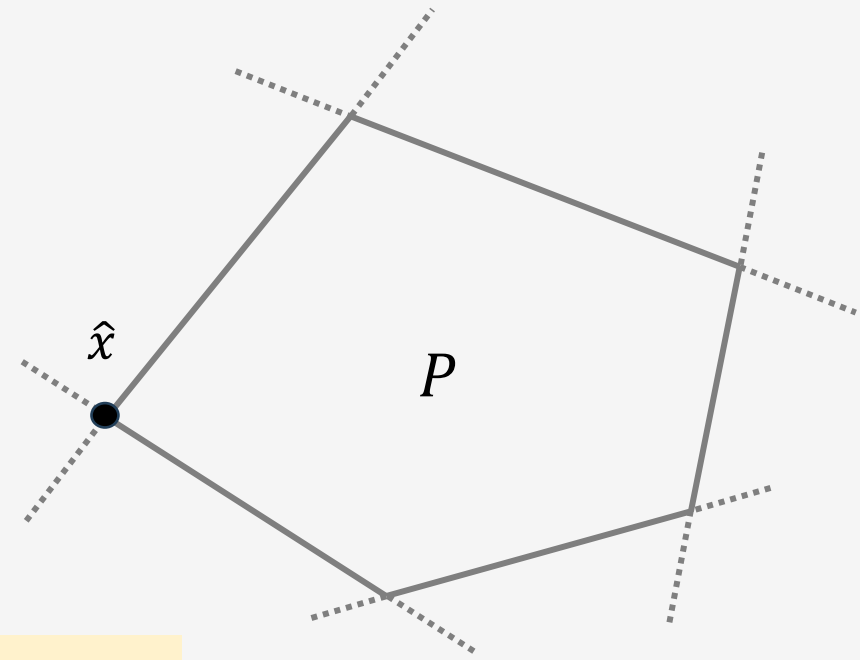
$n = 5$ dimensions

$$P = \{x \in R^n \mid Ax = b, \quad x \geq 0\}$$

Equivalence theorem

Given a nonempty polyhedron

$$P = \{x \in R^n \mid a_i^T x \leq b_i \ i = 1, \dots, m\} \subset R^n$$



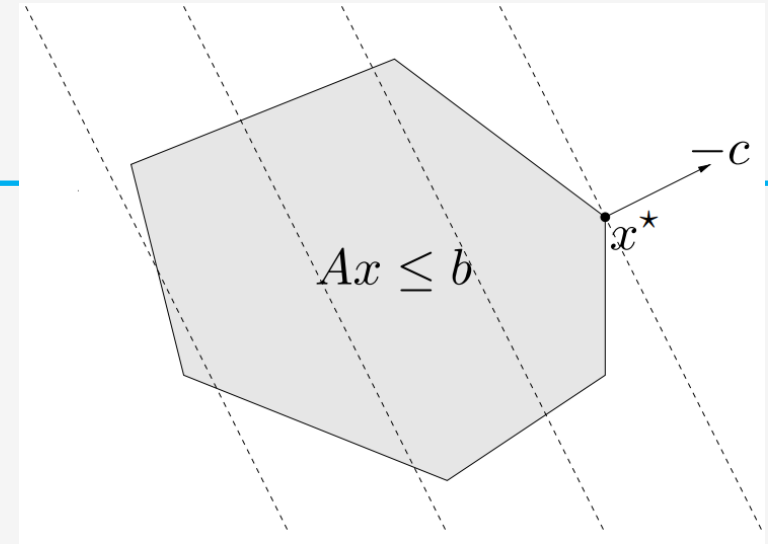
Theorem: if $\hat{x} \in P$ then

$\hat{x}$ is a **vertex** $\Leftrightarrow \hat{x}$ is an **extreme point** $\Leftrightarrow \hat{x}$ is a **basic solution**

Optimality of extreme points

Consider the LP: $\min c^T x$
s.t. $Ax \leq b$

and its feasible region: $P = \{x \in R^n \mid Ax \leq b\}$



Theorem: if P has **at least one extreme** and **there exists an optimal solutions**, then there exists an optimal solution that is an **extreme point** of P .

Solution idea: restrict your search to extreme points only!

How to search among basic feasible solutions

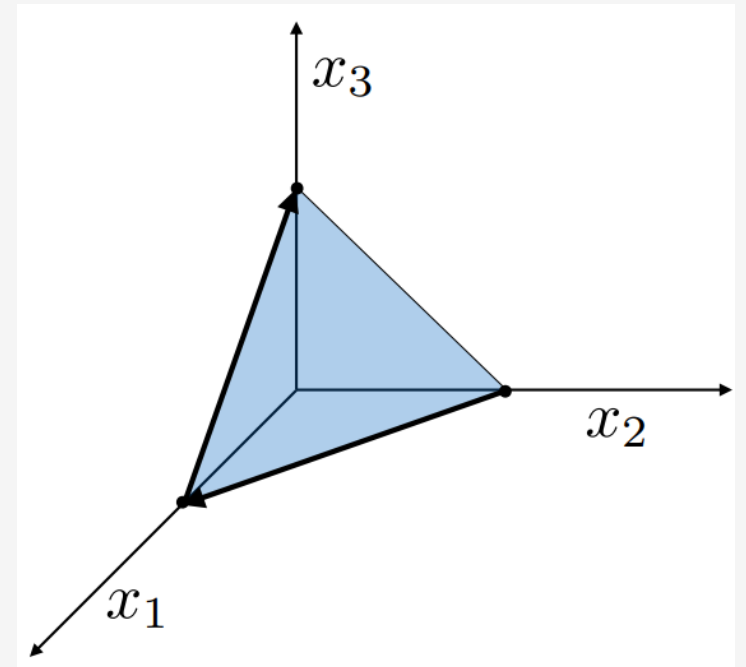
Naïve idea:

Find all basic feasible solutions and select the one that gives the
lowest objective value

It is **intractable** → if $n = 1000$ and $m = 100$ you have to find 10^{143}
solutions!

A more efficient idea (~Simplex method)

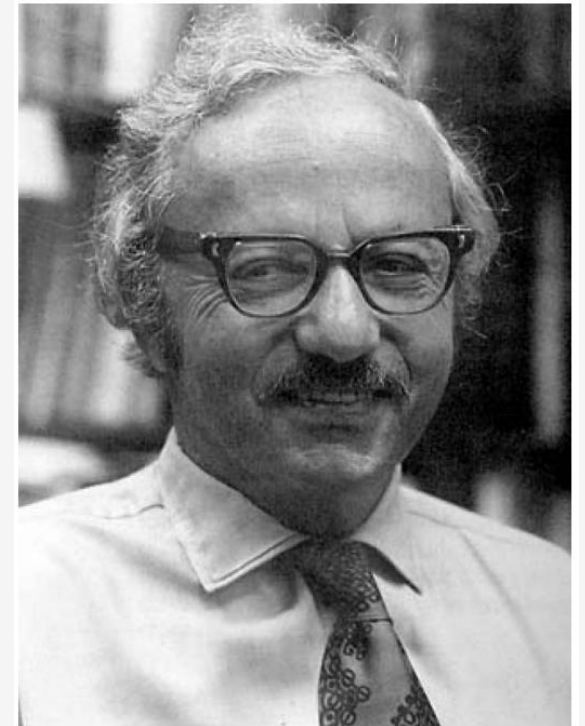
1. **Start** at a corner of the feasible region
2. **Move to** a neighboring corner with lower objective value
3. **Go to** 2, until there is no further improvement



The Simplex method

Among the top 10 algorithms of the 20th

- 1946: Metropolis algorithm
- **1947: Simplex method**
- 1950: Krylov subspace method
- 1951: Decomposition methods for matrix computations
- 1957: Fortran optimized compiler
- 1959: QR algorithm
- 1962: Quicksort
- 1965: Fast Fourier transform
- 1977: Integer relation detection
- 1987: Fast multipole method



George Dantzig developed the Simplex method

Neighboring basic solutions

Neighboring solutions

Two basic solutions are **neighboring** if their basic indices differ by exactly one variable

Very important for the Simplex method!

Example

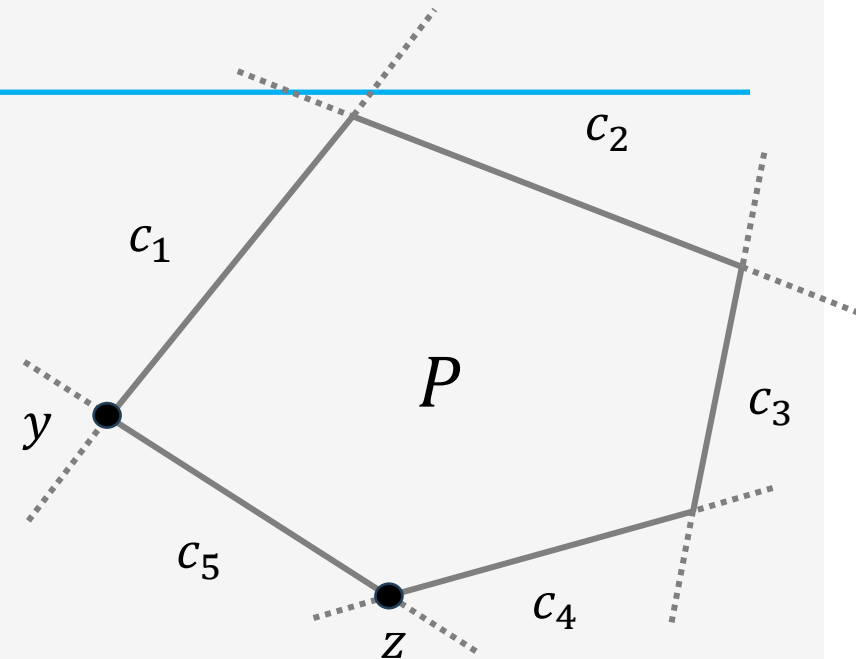
$$Ax = b \Leftrightarrow \begin{bmatrix} 1 & -1 & 0 & 3 & -2 \\ 2 & 0 & -1 & -1 & 0 \\ 0 & 2 & 4 & -1 & 4 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ 14 \end{pmatrix}$$

$$B = \{1, 3, 5\}, \quad x_2 = x_4 = 0$$

$$A_B x_B = b \Rightarrow x_B = A_B^{-1} b \Rightarrow x_B = \begin{pmatrix} x_1 \\ x_3 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2.5 \end{pmatrix}$$

$$\bar{B} = \{1, 3, 4\}, \quad x_2 = x_5 = 0$$

$$A_{\bar{B}} x_{\bar{B}} = b \Rightarrow x_{\bar{B}} = A_{\bar{B}}^{-1} b \Rightarrow x_{\bar{B}} = \begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0.1 \\ 3 \\ -1.7 \end{pmatrix}$$



Feasible directions

Feasible directions

Conditions

Given a basis matrix $A_B = [A_{B(1)}, \dots, A_{B(m)}]$, we can compute a **basic feasible solution**, x , such that:

$$P = \{x \mid Ax = b, x \geq 0\}$$

- x_B solves the system: $A_B x_B = b$
- $x_i = 0, \forall i \neq B(1), \dots, B(m)$

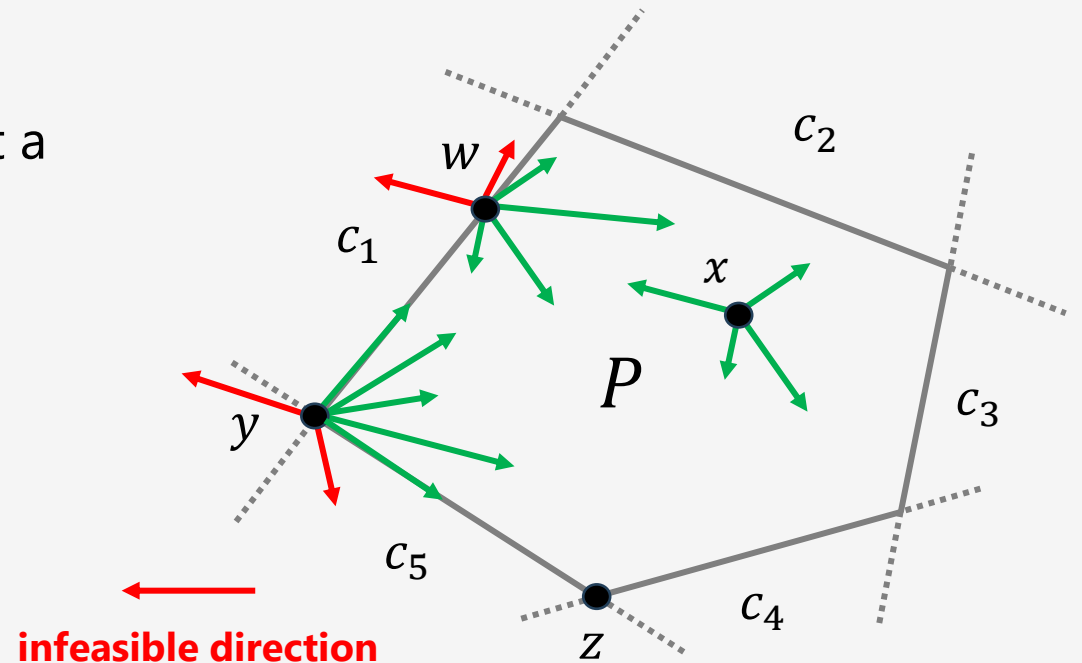
c_i : the i -th constraint of P

- A vector $d \in R^n$ is called a **feasible direction** at a **feasible point** $x \in P$ if

$$\exists \theta > 0 \text{ such that } x + \theta d \in P$$

- For such d and θ we have:

- $A(x + \theta d) = b \Rightarrow \theta A d = 0 \Rightarrow A d = 0$
- $x + \theta d \geq 0$



Feasible directions

Non-basic indices (all i such that $x_i = 0$):

- Pick non-basic index j and set: $d_j = 1 \rightarrow$ (will add j to basis B)
- Set $d_k = 0, \forall k \notin \{j, B(1), \dots, B(m)\}$

Basic indices ($x_B > 0$):

$$Ad = 0 \Rightarrow \sum_{i=1}^n A_i d_i = 0 \Rightarrow A_B d_B + A_j = 0 \Rightarrow A_B d_B = -A_j \Rightarrow d_B = -A_B^{-1} A_j$$

Nonnegativity (non-degenerate assumptions)

- Non-basic variables: $x_i = 0$, and $d_i \geq 0$
- Basic variables: $x_B > 0$, therefore $\exists \theta > 0$ such that $x_B + \theta d_B \geq 0$

$$P = \{x \mid Ax = b, x \geq 0\}$$

Feasible direction d :

- $A(x + \theta d) = b \Rightarrow Ad = 0$
- $x + \theta d \geq 0$

Feasible directions

Example:

$$P = \{x \mid x_1 + x_2 + x_3 = 2, x \geq 0\}$$

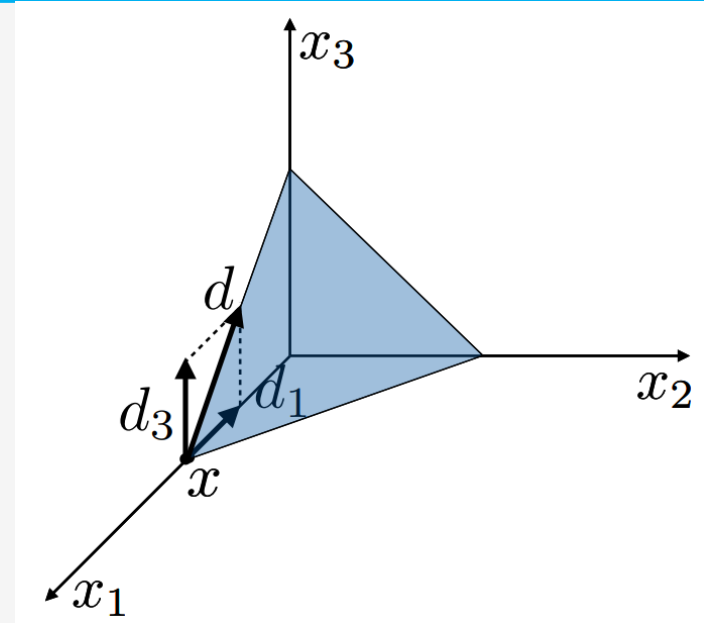
$$x = (2, 0, 0)^T, \quad \underset{\text{Basic idx}}{B = \{1\}} \quad \text{and} \quad \underset{\text{Non-basic idx}}{N = \{2, 3\}}$$

$$\text{New basic index: } j = 3 \rightarrow d_j = 1 \Leftrightarrow d_3 = \mathbf{1}$$

$$\text{Let's find } d_B \rightarrow A_B = [1] \text{ and } A_j = [1]$$

$$A_B d_B = -A_j \Rightarrow d_B = -A_B^{-1} A_j \Rightarrow d_B = -1 \Leftrightarrow d_1 = \mathbf{-1}$$

$$\text{Hence the feasible direction is: } d = (-\mathbf{1}, 0, \mathbf{1})^T$$



How does the cost change

Cost improvement

(for moving towards d)

$$\begin{array}{ccc} & c^T(x + \theta d) - c^T x = \theta c^T d & \\ \nearrow & & \nwarrow \\ \text{New cost} & & \text{Old cost} \end{array}$$

Define the quantity:

$$\bar{c}_j = c^T d = \sum_{j=1}^n c_j d_j = c_j + c_B^T d_B = c_j - c_B^T A_B^{-1} A_j$$

Because $d_j = 1$ and $d_k = 0, \forall k \notin \{j, B(1), \dots, B(m)\}$

We call $\bar{c}_j$ the **reduced cost** of (introducing) variable x_j in the basis

Reduced costs

Interpretation of reduced cost

$$\bar{c}_j = c_j - c_B^T A_B^{-1} A_j$$

Cost per unit increase
of variable x_j

Cost to change other variables,
compensating for x_j to enforce $Ax = b$

- $\bar{c}_j > 0$: adding x_j will increase the objective (**bad**)
- $\bar{c}_j < 0$: adding x_j will decrease the objective (**good**)

Reduced costs for all basic variables is 0

$$\begin{aligned}\bar{c}_{B(i)} &= c_{B(i)} - c_B^T A_B^{-1} A_{B(i)} = c_{B(i)} - c_B^T (A_B^{-1} A_B) e_i = \\ &= c_{B(i)} - c_B^T e_i = c_{B(i)} - c_{B(i)} = 0\end{aligned}$$

Vector of reduced costs

Reduced cost vector (for all variables)

$$\bar{c}_j = c_j - C_B^T A_B^{-1} A_j \Rightarrow \bar{c} = (\bar{c}_1, \bar{c}_2, \dots, \bar{c}_n)^T$$

Isolate the components that are related to basis B

$$\bar{c}_j = c^T d = c_j + c_B^T d_B = c_j - c_B^T A_B^{-1} A_j = c_j - A_j^T (A_B^{-1})^T c_B = c_j - A_j^T p$$

Obtain p by solving the linear system: $A_B^T p = c_B \Rightarrow p = (A_B^{-1})^T c_B$

Computing reduced costs:

- Solve: $A_B^T p = c_B$
- Compute: $\bar{c} = c - A^T p$

Optimality conditions

Optimality conditions

Theorem:

Let x be a **basic feasible solution** associated with basis B , and $\bar{c}$ be the **reduced costs vector**. If $\bar{c} \geq 0$ then x is **optimal solution** for the LP.

Remark:

- This theorem provides a useful **stopping criterion** for the Simplex algorithm
- **Interpretation:** if the neighboring points (solutions) **do not improve the objective** (recall that we minimize $c^T x$) then we should **STOP!**

Optimality conditions

Proof: For a basic feasible solution x with basis B the reduced costs are $\bar{c} \geq 0$

Consider any other feasible solution y and define the direction $d = y - x$

Since x and y are feasible we have: $Ax = Ay = b$ and hence $Ad = 0$. This can be written as:

$$Ad = A_B d_B + \sum_{i \in N} A_i d_i = 0 \Rightarrow d_B = - \sum_{i \in N} A_B^{-1} A_i d_i$$

N is the set of
non-basic variables

The change in objective is:

$$c^T d = c_B^T d_B + \sum_{i \in N} c_i d_i = \sum_{i \in N} (c_i - c_B^T A_B^{-1} A_i) d_i = \sum_{i \in N} \bar{c}_i d_i$$

Since $y \geq 0$ and $x_i = 0, \forall i \in N$, then $d_i = y_i - x_i \geq 0, \forall i \in N$. Then we can write:

$$c^T d = c^T (y - x) \geq 0 \Rightarrow c^T y \geq c^T x$$



Simplex iterations

Stepsize

What happens if for some j we have some $\bar{c}_j < 0$?

We can decrease the cost (objective value) by bringing x_j in the basis

How far can we go?

$$\theta^* = \max\{\theta \mid \theta \geq 0 \text{ and } x + \theta d \geq 0\} \rightarrow d \text{ is the } j\text{-th basic direction}$$

Unbounded:

If $d \geq 0$, then $\theta^* = \infty \rightarrow$ the LP is unbounded

Bounded:

If for some i we have: $d_i < 0$, then
$$\theta^* = \min_{\{i \mid d_i < 0\}} \left(-\frac{x_i}{d_i} \right) = \min_{\{i \in B \mid d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

Note: $d_i = 0, \forall i \notin B$

Moving to a new basis

Next feasible solution

$$x = x + \theta^* d$$

Let $B(\ell) \in \{B(1), \dots, B(m)\}$ be the index such that: $\theta^* = -\frac{x_{B(\ell)}}{d_{B(\ell)}}$

Then: $x_{B(\ell)} + \theta^* d_{B(\ell)} = 0$

New solution:

- $x_{B(\ell)}$ becomes 0 (exists the basis B)
- x_j becomes θ^* (enters the basis B)

New basis

$$A_{\bar{B}} = [A_{B(1)} \quad \dots \quad A_{B(\ell-1)} \quad A_j \quad A_{B(\ell+1)} \quad \dots \quad A_{B(m)}]$$

An iteration of the Simplex method

Initialization:

- A basic feasible solution x
- A basis matrix $A_B = [A_{B(1)} \ \dots \ A_{B(m)}]$

Iterations:

1. Compute the reduced costs $\bar{c}$
 - Solve $A_B^T p = c_B$
 - $\bar{c} = c - A^T p$
2. If $\bar{c} \geq 0$, **STOP**. Declare x **optimal**.
3. Choose j such that $c_j < 0$
4. Compute search direction d with
$$d_j = 1 \quad \text{and} \quad A_B d_B = -A_j$$
5. If $d_B \geq 0$, the problem is **unbounded, STOP**.
6. Compute stepsize $\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$
and set $i = \arg \min_{\{i \in B \mid d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$
7. Define y such that $y = x + \theta^* d$
8. Get new Basis $\bar{B}$ (i exits and j enters)

Example

$$P = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 2, x \geq 0\}$$

$$x = (2, 0, 0)^T \quad B = \{1\}$$

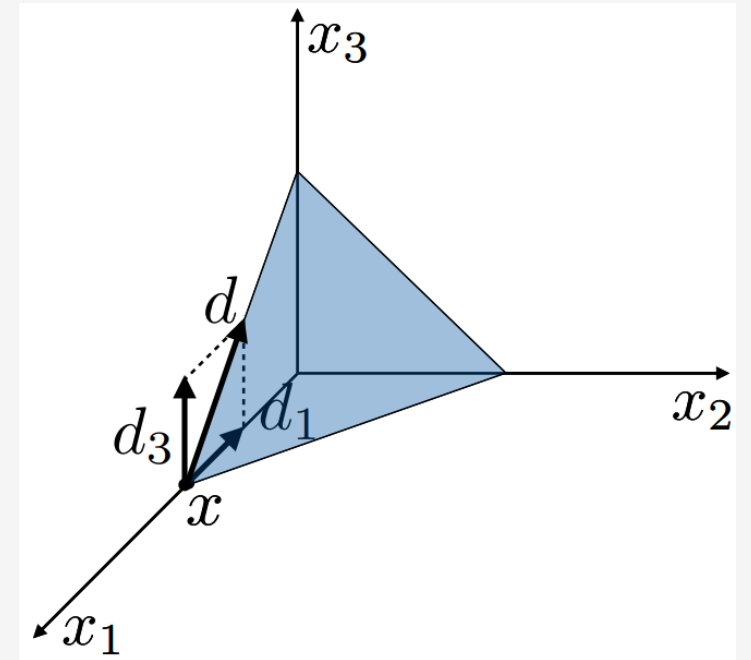
Basic index: $j = 3 \rightarrow d = (-1, 0, 1)^T, \quad d_j = 1$

$$A_B d_B = -A_j \Rightarrow d_B = -1$$

Stepsize:

$$\theta^* = -\frac{x_1}{d_1} = -\frac{2}{-1} = 2$$

New solution: $y = x + \theta^* d = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ **New Basis:** $\bar{B} = \{3\}$



Finite convergence theorem

IF

- $P = \{x \mid Ax = b, x \geq 0\} \neq \emptyset$ (i.e., we have a **feasible LP**) **and**
- Every basic solution is **non-degenerate**

THEN

- The Simplex algorithm **terminates** after a **finite number of iterations**
- At termination we **either** have one of the following results
 - An **optimal basis** B (i.e., the algorithm found the **optimal solution**)
 - A direction d such that $Ad = 0$, $d \geq 0$, $c^T d < 0$ and optimal cost is $-\infty$ (**unbounded**)

Finite convergence theorem

Proof sketch:

At each iteration, the Simplex algorithm improves (reduces) the objective:

- By a positive amount θ^* , along the direction d such that $c^T d < 0$

Therefore, the following two statements hold:

- The cost strictly decreases
- No basic feasible solution can be visited twice

Since there is only a finite number of basic feasible solutions, the Simplex algorithm must eventually terminate. 

References

- Bertsimas/Tsitsiklis: Introduction to Linear Optimization
 - Chapter 3: the Simplex method
- Vanderbei: Linear Programming – Foundations and Extensions
 - Chapter 2: the Simplex method
 - Chapter 6: the Simplex method in matrix notation

Next lecture

- Finding an initial basic solution
- Degeneracy
- Complexity
